

Summary and Recommendations

Overview

This project analyzes customer churn behavior using demographic information, service subscriptions, billing details, contract types, payment methods, tenure, and monthly charges. The objective was to identify the key factors driving customer attrition and provide actionable recommendations to improve customer retention.

The dataset consists of **7,043 customers**, out of which **1,869 customers have churned** while **5,174 customers remain active**.

Overall Churn Rate

- Active Customers: **73.46% (5,174 customers)**
- Churned Customers: **26.54% (1,869 customers)**

This indicates that approximately **1 out of every 4 customers leaves the company**, making churn reduction a critical business priority.

Key Insights

1. Senior Citizens Show Higher Churn Risk

The analysis revealed a significant difference in churn behavior between senior citizens and non-senior citizens.

Customer Type	Retained	Churned
Non-Senior Citizens	76.4%	23.6%
Senior Citizens	58.3%	41.7%

Insight

- Senior citizens churn at a rate of **41.7%**, nearly double that of non-senior customers.
- Although only **1,142 customers** are senior citizens, they represent a highly vulnerable segment.

Business Impact

Special retention programs, dedicated customer support, and simplified service plans should be introduced for senior customers.

2. Customer Tenure Strongly Influences Churn

The tenure distribution clearly shows that churn is concentrated among customers with shorter relationships with the company.

Findings

- Customers with **0–12 months tenure** show the highest churn.
- Churn gradually decreases as tenure increases.
- Customers with **60–72 months tenure** demonstrate the highest retention levels.

Insight

The first year of the customer lifecycle is the most critical period for retention.

Recommendation

Implement onboarding campaigns and loyalty rewards during the first 12 months.

3. Contract Type is the Strongest Predictor of Churn

Contract analysis revealed dramatic differences in churn behavior.

Contract Type	Retained	Churned
Month-to-Month	2,220	1,655
One Year	1,307	166
Two Year	1,647	48

Churn Rates

- Month-to-Month: **42.7% churn**
- One-Year Contract: **11.3% churn**
- Two-Year Contract: **2.8% churn**

Insight

Customers on month-to-month contracts are approximately **15 times more likely to churn** than customers on two-year contracts.

Recommendation

Encourage customers to migrate to annual and long-term contracts through discounts and loyalty incentives.

4. Paperless Billing Customers Churn More Frequently

Paperless Billing	Retained	Churned
Yes	2,771	1,400
No	2,403	469

Churn Rates

- Paperless Billing Users: **33.6%**
- Non-Paperless Billing Users: **16.3%**

Insight

Customers using paperless billing exhibit more than twice the churn rate compared to customers using traditional billing.

Recommendation

Investigate whether paperless billing customers are receiving insufficient engagement or communication.

5. Payment Method Significantly Affects Retention

Payment Method	Churned Customers
Electronic Check	1,071
Mailed Check	308
Bank Transfer (Automatic)	258
Credit Card (Automatic)	232

Churn Rates

- Electronic Check: **45.3%**
- Mailed Check: **19.1%**
- Bank Transfer Automatic: **16.7%**
- Credit Card Automatic: **15.2%**

Insight

Customers paying through Electronic Check have the highest churn risk.

Recommendation

Promote automatic payment options through incentives and reminders.

6. Monthly Charges Influence Churn

The boxplot analysis shows a clear relationship between monthly charges and churn.

Findings

- Churned customers have a median monthly charge of approximately **₹80–85**.
- Retained customers have a median monthly charge of approximately **₹65**.

Insight

Customers paying higher monthly fees are more likely to leave.

Recommendation

Review pricing strategies and offer customized packages for high-paying customers.

7. Service Adoption Reduces Churn

Several service-based analyses reveal that customers who use additional services are less likely to churn.

High Churn Observed When Customers Lack:

- Online Security
- Online Backup
- Device Protection
- Tech Support

Example

Customers without Online Security or Tech Support show substantially higher churn counts compared to customers who subscribe to these services.

Business Insight

Value-added services increase customer stickiness and strengthen retention.

Recommendation

Bundle support, backup, and security services into subscription packages.

8. Internet Service Type Affects Churn

Findings

Customers using **Fiber Optic Internet** exhibit significantly higher churn compared to DSL users.

Possible Reasons

- Higher pricing
- Increased customer expectations
- Service quality concerns

Recommendation

Conduct customer satisfaction surveys among Fiber Optic users.

9. Family Status Influences Customer Retention

Partner Analysis

Customers without partners churn noticeably more than customers with partners.

Dependents Analysis

Customers with dependents show substantially lower churn rates.

Insight

Customers with stronger family commitments tend to maintain long-term subscriptions.

10. Gender Has Minimal Impact

The gender analysis shows nearly identical churn patterns among male and female customers.

Insight

Gender is not a significant predictor of customer churn in this dataset.

Advanced 3D Analysis

A 3D scatter plot was created using:

- Tenure
- Monthly Charges
- Churn Status

Findings

The highest concentration of churned customers appears among:

- Low Tenure Customers
- High Monthly Charges Customers

Customers with:

- Long tenure
- Stable subscriptions

are significantly more likely to remain with the company.

High-Risk Customer Profile

A customer is most likely to churn when they have:

- ✓ Month-to-month contract
- ✓ Electronic check payment method
- ✓ High monthly charges
- ✓ Low tenure
- ✓ No online security services
- ✓ No tech support services
- ✓ Senior citizen status

Business Recommendations

1. Convert month-to-month customers into annual plans.
2. Offer incentives for automatic payment methods.
3. Focus retention campaigns during the first year of customer tenure.
4. Bundle Online Security, Backup, and Tech Support services.
5. Develop targeted retention programs for senior citizens.
6. Review pricing strategies for customers with high monthly charges.
7. Investigate churn causes among Fiber Optic users.
8. Build a churn prediction model to proactively identify at-risk customers.

Conclusion

The analysis identified **contract type, tenure, payment method, monthly charges, senior citizen status, and service adoption** as the strongest drivers of customer churn. With an overall churn rate of **26.54%**, the company can significantly improve customer retention by focusing on high-risk segments, particularly customers on month-to-month contracts, electronic check users, senior citizens, and customers with limited service engagement. Implementing targeted retention strategies for these groups can reduce churn, increase customer lifetime value, and improve long-term business profitability.